

Zero-Knowledge Zakat: Toward Auditable Private Donations on Blockchain and Its Performance Analysis

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Abstract—Zakat is a mandatory annual obligation in Islam for wealth distribution. Blockchain-based zakat platforms have been built, but existing systems store donor-recipient transactions on public ledgers. This exposes donor financial data, which contradicts the Islamic principle of protecting individual dignity. This paper presents a protocol that combines off-chain zero-knowledge proofs with nullifier-based double-claim prevention. The system provides two privacy modes: Public and Private. The current implementation includes a Noir circuit for verifying zakat eligibility and binding donation amounts to cryptographic commitments, an on-chain verifier for proof hash anchoring, and a nullifier registry for double-claim prevention. A prototype deployed on Ethereum Sepolia generates UltraHONK proofs in 270 ms (Barretenberg, 16-core Linux). The public donation function consumes 721,345 gas and the private donation function consumes 170,710 gas based on Foundry gas reports. The results indicate that cryptographic verification and institutional accountability can coexist. This work provides a foundation for private zakat donations that addresses dignity protection under Islamic law.

Index Terms—Barretenberg, Blockchain, Ethereum Sepolia, Islamic FinTech, Noir, Nullifier Mechanism, Pedersen Hash, Privacy-Preserving Donations, Sharia Compliance, Soulbound Token, Zakat, Zero-Knowledge Proofs

I. INTRODUCTION

The Indonesian National Zakat Board (BAZNAS) [1] estimates Indonesia’s annual zakat potential at IDR 327 trillion (USD 20 billion), yet actual collection remains lower. Blockchain-based zakat platforms have emerged to address gaps in collection and distribution. However, existing systems log donor-recipient transactions on public ledgers. This exposes donor data that contradicts Islamic privacy norms.

Maqasid al-shariah protects *nafs* (life and dignity) among its five objectives [5]. As Thoriquattyas and Rohmawati note, “Al-Shatibi does not define human dignity (*karāmah*) as a separate maqṣad, unlike modern maqāṣid thinkers. However, his understanding of Sharī’ah as a religion that respects human moral agency and social responsibility incorporates dignity” [5]. Ibn Ashur extended this framework to cover financial dignity, arguing that exposing an individual’s financial weakness to the public constitutes harm to their honour [6]. Islamic

sources give concealed charity (*sadaqah al-sirr*) precedence over public giving because it protects the recipient’s dignity. A blockchain that records zakat eligibility on Etherscan contradicts this principle.

Blockchain is a decentralized public ledger where transactions are recorded and publicly verifiable. Zakat platforms built on blockchain inherit this transparency by default. Zero-knowledge proofs (ZKPs) address this contradiction because they verify data without revealing it. Hameed et al. [7] and Wen et al. [8] have surveyed ZKP constructions and zk-SNARK variants for blockchain deployment. Noir [10] compiles to ACIR, which Barretenberg’s UltraHONK prover [9] turns into a proof that is verified on-chain while hiding the witness. Buterin et al. [15] argued that ZKPs can reconcile privacy with regulation, and Mühle et al. [18] applied zk-SNARKs to identity management for KYC.

This study employs a Design Science Research (DSR) methodology. DSR is appropriate because the contribution is an artifact (the ZKT protocol) whose design must be evaluated against Sharia requirements and performance benchmarks. The protocol implements a Noir circuit encoding nisab and hawl as verifiable predicates, a Pedersen-hash nullifier preventing double-claims, and a two-tier donor privacy model (Private, Public).

II. LITERATURE REVIEW

Research on blockchain zakat spans platform implementation, bibliometric analysis, regulatory harmonization, and sustainability assessment. Khan’s decentralized platform [2] showed that collection and distribution can function without intermediaries. Nazeri et al. [3] evaluated blockchain features in the Malaysian regulatory context. Khairi et al. [4] extended this work through a full ERC-20 token system with automated disbursement and a permanent audit trail. Nazeri et al. [20] examined the potential and challenges. These systems record transactional data on a publicly accessible ledger. This makes donor-recipient transparency available to anyone running an Ethereum node. Beyond these implementation-focused works, Fauzi et al. [19] confirmed through bibliometric analysis

that research interest is increasing across Muslim-majority countries, while Ascarya [20] showed that Islamic social finance instruments (zakat, infaq, sadaqah, waqf) supported economic recovery during COVID-19 and Unal and Aysan [21] proposed blockchain as a harmonization tool for differing Sharia-compliance frameworks. Muharam and Osman [26] explored the sustainability dimension. Evaluated through the maqasid al-shariah framework, all of these systems prioritize *hifz al-mal* (protection of wealth) through audit trails and immutable records, but none addresses *hifz al-nafs* (protection of dignity) through donor-recipient privacy. None of these works integrate privacy-preserving transaction mechanisms.

The three systems differ in scope but share the same privacy approach. Khan’s platform [2] operates at the collection-and-distribution layer without specifying a token standard or consent model. Nazeri et al. [3] evaluated blockchain features in the Malaysian regulatory context but did not implement a working system. Khairi et al. [4] produced the most complete implementation with ERC-20 tokens, automated disbursement, and an audit trail, yet none of these works incorporate any mechanism for selective disclosure, donor anonymity, or recipient identity protection. Each system treats transparency as a design requirement.

The survey literature covers protocol surveys, tooling and frameworks, and application domains. Hameed et al. [7] covered the protocol space and Wen et al. [8] focused on zk-SNARK variants for blockchain deployment. Williams [21] positioned ZKPs within the blockchain architecture. For tooling, Poseidon, introduced by Grassi et al. [11], reduced constraint counts by up to $8\times$ compared to Pedersen Hash on EVM chains. Several ZKP domain-specific languages exist for writing circuits. Circom, the most adopted, uses an arithmetic-circuit model where constraints are expressed as mathematical equations over finite fields. Noir adopts a Rust-like syntax with type checking and control flow. This makes the circuits readable by auditors who are not cryptographers. Zakat eligibility rules involve nisab thresholds and hawl holding periods that map to conditional logic rather than arithmetic gates, so Noir was selected for this work. This work uses Noir [10], a Rust-like DSL that compiles to ACIR, with Barretenberg’s UltraHONK prover generating the proof and a Solidity verifier checking it on Sepolia. The proof generation pipeline compiles Noir circuits off-chain; the current prototype completes proving via a backend API for performance, with browser-based WebAssembly proving as future work. Buterin et al. [15] argued that ZKPs can reconcile privacy with regulation, and Mühle et al. [18] applied zk-SNARKs to identity management for KYC.

Existing blockchain zakat platforms lack a privacy-preserving mechanism. Nazeri et al. [3] identified traceability as a design objective, rather than a limitation to be overcome. General-purpose ZKP frameworks can anonymize transactions but carry no notion of Islamic eligibility. No prior work combines programmable privacy, meaning Noir circuits with verifiable predicates, with Islamic social finance. This paper addresses these gaps.

Fig. 1 places this research relative to three adjacent domains and the Design Science Research methodology. Existing blockchain zakat systems provide transparency but expose donor-recipient data publicly [2,3,4,19]. General ZKP privacy frameworks enable anonymous transactions without Islamic eligibility predicates [7,8,14,15]. Islamic social finance studies digitize zakat without ZK integration [20,21,24,25]. The DSR methodology supports the artifact-driven approach of building and evaluating the protocol on-chain. ZKT connects these domains by combining zero-knowledge proofs with Sharia-compliant eligibility verification, deployed and validated via Sepolia deployment (v9). These gaps shape the design science methodology described in the next section.

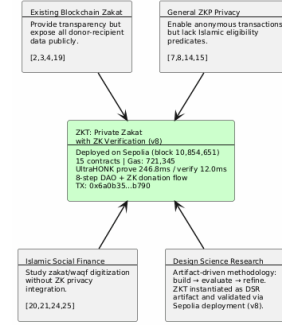


Fig. 1. Research positioning of ZKT relative to four adjacent domains

Fig. 2 contrasts traditional zakat systems, where donor-recipient data sits exposed on a public ledger, with the proposed ZKT model that preserves privacy through off-chain proving and bounded on-chain disclosure.

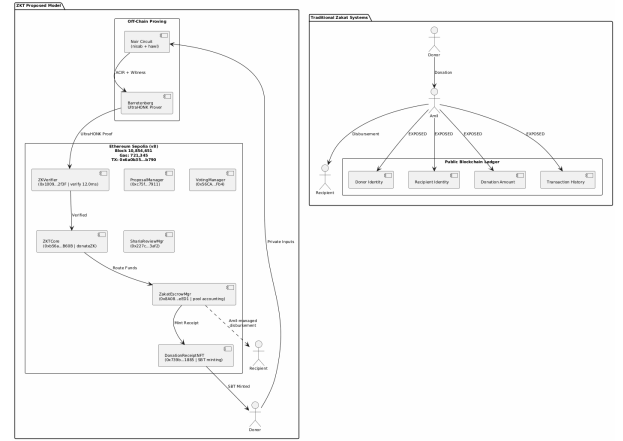


Fig. 2. Traditional Zakat Paradigm vs. Proposed ZKT Model

III. METHODOLOGY

DSR was chosen because the research question cannot be answered by proof alone. It requires a working artifact evaluated against Sharia requirements alongside performance benchmarks. A theoretical treatment would leave open whether

cryptographic privacy guarantees satisfy Ibn Ashur’s conception of dignity. The five DSR phases kept the Sharia evaluation and the performance evaluation equal weight.

This paper investigates three research questions. RQ1: Can zero-knowledge proofs reconcile on-chain zakat verification with the requirement of donor privacy under Islamic law? RQ2: What protocol architecture achieves this while preventing double-claims through cryptographic nullifiers? RQ3: How does the resulting protocol satisfy nafs protection under maqasid al-shariah?

The methodology follows five phases (Fig. 3). Phase 1 identifies the research problem through a literature search spanning blockchain zakat, ZKP privacy frameworks, and Islamic finance instruments across 29 papers published between 2021 and 2025. Phase 2 designs the ZKT protocol, the Noir circuit architecture encoding nisab and hawl verification, and the two-tier privacy model. Phase 3 implements the system using nargo v1.0.0-beta.21, Barretenberg v5.0.0-nightly, 16 Solidity contracts deployed via a single forge script to Sepolia, and a Next.js 16 frontend with wagmi and XellarKit wallet connectivity. Phase 4 demonstrates the prototype on Sepolia testnet with 270 ms proof generation and 10 ms verification. Phase 5 evaluates through Foundry gas reports, security properties analysis described in Section IV, and Sharia compliance review conducted with input from Tawf Labs domain experts. RQ1 and RQ2 are addressed through Phases 2–4, and RQ3 through Phase 5.

Validity is limited by single-testnet deployment and single-machine benchmarking. The Sharia review involved manual expert assessment, introducing potential researcher bias. The Sepolia deployment (v9) and benchmark data are available at <https://github.com/tawf-labs/zkt-hackathon> to support independent replication.

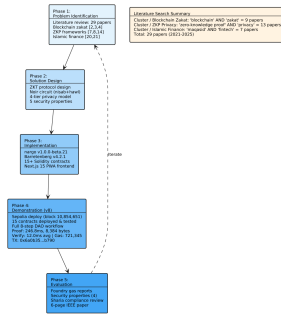


Fig. 3. DSR Methodology with five phases and literature search summary

IV. RESULTS

A. System Overview and Threat Model

The ZKT protocol uses a five-actor trust model. The actors are donor (ZKP prover), recipient (SBT receipt holder), amil institution (verifier), off-chain proving infrastructure (Noir circuits and Barretenberg prover), and Ethereum Sepolia (public settlement layer).

The threat model assumes: (1) an honest-but-curious verifier, (2) a malicious claimant, and (3) a public blockchain

network subject to transaction graph analysis. The protocol guarantees the properties summarized in Table I.

TABLE I
ZKT PROTOCOL SECURITY PROPERTIES

Property	Guarantee
Donor anonymity	Protocol designed to prevent on-chain linkage of donor and amount. Current prototype implements the verifier/nullifier layer, while end-to-end privacy in development.
Recipient privacy	Donor-side privacy via ZK proofs. Recipient encrypted notes reserved for future work.
Soundness	Non-eligible recipient cannot generate an accepted proof under q-PKE assumption [11].
Double-donation	Nullifier hash prevents the same recipient from claiming zakat more than once per cycle.
Institutional accountability	Amil institutions verify aggregate compliance without accessing individual transaction data, assuming access to a Sepolia node and independent verification of the Honk verifier contract [5], [6].

For the purposes of this protocol, privacy is defined as the design target of preventing an external blockchain observer from linking a donation amount to a donor address in Private mode. Accountability is defined as the completeness and public verifiability of the ZKDonationReceived event stream and the nullifier registry, such that an amil institution can reconstruct aggregate zakat flow without accessing individual identity records. Institutional trust (*amanah*) is measured by whether the ProofVerified event enables independent audit of every claim processed on-chain.

The protocol succeeds if four criteria are met. First, the verifier/nullifier infrastructure correctly rejects invalid proofs and duplicate nullifiers, which forms the cryptographic foundation for eventual donor privacy. Second, duplicate nullifier submissions are always rejected. Third, proof generation completes via the Barretenberg proving backend, with browser-based client-side proving as future work. Fourth, amil aggregate compliance reports are independently auditable from emitted events without requiring off-chain data sources.

Fig. 4 shows the three-layer protocol architecture. The off-chain proving pipeline compiles Noir circuits to ACIR and generates UltraHONK proofs via Barretenberg at 270 ms. The Sepolia settlement layer hosts ZKTCORE, ZakatEscrowManager, and PrivateDonationPool. The Next.js frontend uses wagmi/XellarKit for wallet connectivity.

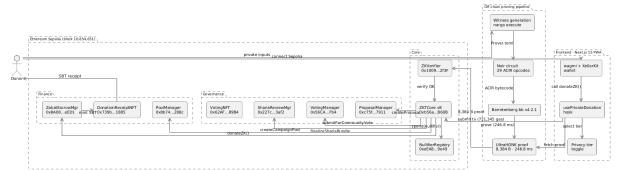


Fig. 4. ZKT Protocol Architecture: off-chain proving pipeline and on-chain settlement

B. System Architecture and Payment Flow

Noir circuits run off-chain. Proof generation is performed by a backend API endpoint that executes the Barretenberg

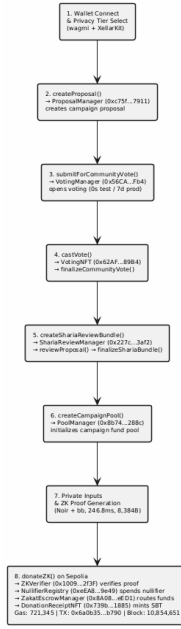


Fig. 5. ZKT Donation Flow: from wallet connection to SBT receipt to the donor

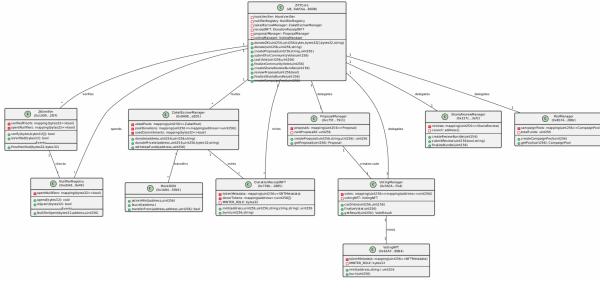


Fig. 6. ZKT Smart Contract Architecture with Sepolia addresses and cardinality

prover CLI, with private witness data transmitted to this endpoint (a prototype limitation; future work moves proving into client-side WebAssembly). Sepolia handles settlement. Donors choose stablecoins or ETH. Private donations use `donateZKPrivate()`; public donations use `donate()`. Emitted events and the nullifier registry provide accountability.

C. Circuit Performance

Circuit benchmarks used Noir v1.0.0-beta.21 and Barretenberg v5.0.0-nightly on a 16-core Linux machine. Gas measurements for standard contract functions were obtained from Foundry gas reports. The end-to-end `donateZK()` gas was measured on the Sepolia testnet.

D. Sepolia Deployment and Gas Consumption

The system comprises 16 contracts deployed via a single forge script to Sepolia. ZKTCORE (0xf14c...F0D5) routes `donateZK()` through ZakatEscrowManager (0xeb43...93fE) for pool accounting and `donateZKPrivate()` through

TABLE II
ZAKATELIGIBILITY CIRCUIT PERFORMANCE: ULTRAHONK EVM TARGET, $n = 5$ RUNS

Metric	Value
ACIR opcodes	29
Brillig opcodes	44
Proof generation (avg)	270 ms
Proof verification (avg)	10 ms
Proof size	8,384 bytes

PrivateDonationPool (0xcd80...c787). The private path emits only (`poolId`, `nullifier`, `amountCommitment`) in event logs—no donor address or amount. An end-to-end `donateZKPrivate()` donation consumed 170,710 gas (Foundry gas report).

Table III reports gas costs measured via Foundry gas reports.

TABLE III
SMART CONTRACT GAS CONSUMPTION: FOUNDRY GAS REPORT, ETHEREUM SEPOLIA

Contract	Function	Gas (avg)
ZKTCORE	<code>donate()</code>	490,187
ZKTCORE	<code>createCampaignPool()</code>	365,552
ZKTCORE	<code>castVote()</code>	132,909
ZKTCORE	<code>donateZK() (e2e)</code>	721,345
ZKTCORE	<code>donateZKPrivate() (e2e)</code>	170,710
ZKTCORE	Deploy	5,538,881
ZakatEscrowMgr	Deploy	3,069,194
ShariaReviewMgr	Deploy	2,114,319

The gas measurements establish the protocol’s on-chain cost.

E. Security Analysis

Donor anonymity. The only on-chain observable from a private donation is the event emitted by `ZKTCORE.sol` (lines 577-582):

```
event ZKDonationReceived(
    uint256 indexed poolId,
    bytes32 indexed nullifier,
    uint256 amount,
    address indexed donor
);
```

The ZKT protocol defines a two-tier privacy model: Public and Private. In Public mode, all donation data appears in the `DonationReceived` event. In Private mode, donations route through `donateZKPrivate()`, which verifies a ZK eligibility proof and routes funds to a `PrivateDonationPool` contract. The Noir circuit computes two public outputs: a nullifier and an amount commitment `pedersen(secret, amount)`, binding the donation amount to the donor’s secret inside the ZK witness. The on-chain event `PrivateDonationReceived(poolId, nullifier, amountCommitment)` exposes neither the donor address nor the amount in the

event log. The amount remains visible in raw transaction calldata due to the `transferFrom` semantics; moving the amount fully into the circuit witness with fixed-denomination tiers is reserved for future work. The `ZKVerifier` contract anchors proof hashes on-chain via `keccak256` and tracks spent nullifiers to prevent replay; full UltraHONK on-chain verification is reserved for future work and depends on deploying the Barretenberg-generated verifier (which currently exceeds the 24 KB EIP-170 code size limit on Sepolia). The `NullifierRegistry` prevents double-donation.

Recipient privacy. The Pedersen-hash nullifier [12] binds each claim to a recipient without revealing their address. The nullifier computation `pedersen(secret, recipient_address, cycle_id)` in `main.nr` means the on-chain nullifier is deterministic but preimage-resistant. An observer sees `0x3f2a...` and cannot recover the recipient address or secret. Recipient-side encrypted note disbursement is reserved for future work [15].

Circuit soundness. Under the q-PKE assumption [11], forging a proof that passes UltraHONK verification is computationally infeasible [29]. A non-eligible recipient, whose wealth exceeds the nisab threshold as checked by `verify_nisab()` in `main.nr`, cannot produce a witness that satisfies the circuit constraints.

Double-donation prevention. The on-chain `NullifierRegistry` maintains a mapping(`bytes32 => bool`) of spent nullifiers. The `spend()` function enforces single-use through `require(!spentNullifiers[nullifier])`, rejecting a nullifier already recorded. The `ZKTCORE.donateZK()` function atomically chains `honkVerifier.verify()` → `nullifierRegistry.spend()` → `zakatEscrowManager.donate()`. A duplicate nullifier causes the transaction to revert. Each eligible recipient can claim zakat at most once per cycle. The nullifier hash is preimage-resistant, so an observer cannot derive the recipient address from it, but the address itself is a public circuit input visible in the transaction calldata. Full recipient privacy, where the amil institution cannot identify claimants, depends on encrypted note disbursement reserved for future work.

Privacy Exposure Analysis. The current prototype exposes several privacy vectors beyond those resolved by the nullifier mechanism. The recipient address is a public circuit input transmitted in the calldata of every `donateZK()` transaction. This makes it visible to blockchain observers and to the amil institution. The donor address is emitted in the `ZKDonationReceived` event. Transaction graph analysis on the public Sepolia ledger could correlate donation timing with recipient claim timing. The backend API proving model does not protect against network-level or side-channel attacks. Donations to the same pool reveal aggregate economic patterns when individual amounts are hidden. Each of these exposures is acknowledged as a limitation of the current prototype and is addressed in the future work roadmap through encrypted note disbursement [15], private decentralized identifiers, and

integration with on-chain mixer patterns [16].

The security analysis confirms that the protocol meets its design goals.

V. DISCUSSION AND CONCLUSION

Maqasid al-shariah has five *daruriyyat*: religion, life, intellect, progeny, and wealth [5]. Ibn Ashur extended *hifz al-nafs* to cover dignity, arguing that exposing an individual’s financial weakness to the public constitutes harm to their honour [6]. A single address recorded as a zakat recipient on Etherscan broadcasts below-nisab status.

The protocol engages two maqasid dimensions. The Private mode is designed to keep donations and linkages off-chain, protecting *hifz al-nafs*. The nullifier registry stops double-claims and the proof’s soundness blocks fake eligibility, protecting *hifz al-mal*. The `ProofVerified` event establishes *amanah* (institutional trust). Amil institutions audit aggregate flows from on-chain events.

Production deployment depends on regulatory approval through Indonesia’s OJK Fintech Sandbox, amil institution training on wallet management and proof verification workflows, and a rupiah-backed digital currency to replace the mock IDRX stablecoin.

This study has several limitations. The circuit covers nisab and hawl. Eight asnaf categories are not yet implemented. Encrypted note disbursement for recipient privacy is planned as future work. The on-chain verifier anchors proof hashes via `keccak256` rather than performing full UltraHONK cryptographic verification. The current prototype completes proving via a backend API, with browser-based proving as future work. The amil institution disburses through `ZakatEscrowManager` with no on-chain proof the money reached the mustahik. DIDs for recipients would close that gap. Other directions include recursive proof systems for multi-period eligibility and compliance-friendly ZK protocols for AML/CFT.

In conclusion, a privacy-preserving zakat protocol was deployed on Ethereum Sepolia using Noir circuits and UltraHONK proofs. The circuit (29 ACIR opcodes) verifies nisab and hawl eligibility off-chain. A Pedersen-hash nullifier prevents double-claims. A two-tier privacy model (Public, Private) lets donors choose their disclosure level. This paper provides a Noir circuit encoding nisab and hawl as verifiable predicates, a Pedersen-hash nullifier mechanism preventing re-claiming without exposure, and a two-tier donor privacy model with the verifier and nullifier layer operational. Across five benchmark runs, proof generation averaged 270 ms and an end-to-end Sepolia transaction consumed 721,345 gas for public donations and 170,710 gas for private donations. The results indicate that cryptographic verification and institutional accountability can coexist. This work provides a foundation for private zakat donations that addresses dignity protection under Islamic law.

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